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EDITORIAL REVIEW

Plasma modelling and numerical simulation

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Abstract

Plasma modelling is an exciting subject in which virtually all physical disciplines are represented. Plasma models combine the electromagnetic, statistical and fluid dynamical theories that have their roots in the 19th century with the modern insights concerning the structure of matter that were developed throughout the 20th century.

The present cluster issue consists of 20 invited contributions, which are representative of the state of the art in plasma modelling and numerical simulation. These contributions provide an in-depth discussion of the major theories and modelling and simulation strategies, and their applications to contemporary plasma-based technologies. In this editorial review, we introduce and complement those papers by providing a bird's eye perspective on plasma modelling and discussing the historical context in which it has surfaced.

1. Plasma modelling and numerical simulation

Since its inception at the beginning of the 20th century, low-temperature plasma science has grown into a major field of science. Low-temperature plasma sources and gas discharges are found in domestic, industrial, atmospheric and extra-terrestrial settings. Examples of domestic discharges are those used in lighting [1] and plasma display panels (PDPs) [2]. In industry, plasma sources are widely used for material processing; among the many applications are deposition and feature etching in the fabrication of electronic devices [3] and surface modification of textiles [4]. More recently, biomedical applications of plasmas have been developed [5].

In order to improve the performance of scientific and technological plasma sources, a thorough understanding of such devices is required. Traditionally, this goal has been achieved by carrying out *experiments and modelling*. In the last few decades, *numerical simulation* has become an indispensable addition to plasma diagnostics, especially in situations where direct measurements are costly or difficult to implement. These complications may be due to the limited accessibility to the plasma, due to small length or time scales or

simply because the quantities of interest are difficult to measure directly with any available measurement technique.

In early numerical plasma simulation codes, gross simplifications of the actual plasma sources were necessary: such calculations were typically restricted to one-dimensional geometries, and based on simplified physical models. As an example, low-pressure gas discharge modelling has long been based on the assumption that no heating of the background gas takes place (e.g. [6]), which can be shown to be false under various circumstances [7–9]. Likewise, the deviations from thermodynamic equilibrium in high-pressure plasmas have been disregarded [10] long after they had been established [11].

In recent years, self-consistent three-dimensional numerical codes with few concessions to the physical model have started to appear, at least for some plasma regimes. The complexity that is involved in such codes suggests that the code is constructed in such a way that it—or parts of it—can be re-used for other applications than the one for which it had originally been developed. In *toolkit designs*, the code is based on reusable building blocks that implement sub-models that can be assembled to form a particular self-consistent physical model. Examples of codes that have been developed along these lines

are the hybrid plasma equipment model (HPEM) toolkit of Kushner and co-workers [12, 13] and the Plasimo modelling framework that is being developed at Eindhoven University of Technology [14, 15]. In recent years, also some general-purpose commercial multi-physics codes have become sufficiently capable of simulating various types of non-equilibrium plasmas self-consistently; as an example, this progress is demonstrated by the increased level of sophistication in the Comsol-based studies of the plasma needle by Sakiyama in the group of Graves at Berkeley [16–20].

In this special issue of the *Journal of Physics D: Applied Physics*, some of the leading workers in the field of plasma modelling and simulation present recent developments in their respective fields of expertise. Our goal has been to cover the major contemporary modelling techniques in a collection of review-style papers. These are accompanied by a series of papers in which these techniques are deployed in order to simulate particular applications or application groups. We realize that such an overview can never be complete and acknowledge that, in particular in the selection of application-oriented subjects, other choices would have been equally valid. Nevertheless, we are confident that the result gives a good impression of the major trends in contemporary plasma modelling and numerical simulation.

In this paper we will introduce the 20 contributions to this special issue. We will do so by providing the historical context in which the key scientific concepts came into existence. Firstly, this ensures that appropriate credit is given to the scientists who developed plasma and gas discharge science throughout the 20th century (section 2). Secondly, such an exposition allows us to point out the important events in the development of plasma modelling and simulation in particular.

This introduction has been organized around the three main types of plasma models. The statistical methods for the evaluation of transport in non-equilibrium plasma are considered in section 3. In section 4, we provide an overview of the key concepts in modelling plasmas in or near local thermal equilibrium (LTE). We continue with a discussion of particle and hybrid modelling techniques (section 5) and make some concluding remarks in section 6.

2. The theoretical foundation of plasma modelling

Let us first consider the various constituents of a self-consistent plasma model. It typically contains a combination of the following sub-models:

- A module for calculating the electromagnetic field in the plasma region and the other regions that form the plasma system is needed. This electromagnetic field is governed by the Maxwell equations (dating from the year 1865) [21].
- Depicting the plasma as a single fluid, a module for calculating its barycentric velocity $\vec{v}$, mass density ρ and pressure p is required. In general, these are governed by the momentum balance equations; since a plasma is a Newtonian fluid, these reduce to the Navier–Stokes equations (1822) [22, 23].

- The Navier–Stokes equations must be closed by an equation of state, which expresses the relation between the mass density and pressure field. This is usually obtained by combining Dalton’s law $p = \sum_i p_i$ (1805) [24] for the (total) pressure p with the ideal gas law $p_i = nkT_i$ (1834) [25] for the partial pressures p_i of the species i in the plasma. Here k is Boltzmann’s constant, n_i and T_i are the species’ particle density and temperature, respectively.
- The calculation of the density n_i , temperature T_i and the velocity $\vec{u}_i$ of species i is in the realm of a *plasma composition module*.

In the case of systems in LTE, the densities n_i are functions of the temperatures T_i and partial pressures alone and can be obtained from the law of mass action (1864), which has its roots in the works of Guldberg, Waage and Van’t Hoff [26–29].

The temperatures are governed by heat balance equations. These depend on a heat flux, which is due to heat *advection* and *conduction*. The latter concept was modelled by Joseph Fourier (1822) [30, 31, section VIII, item 98].

When deviations from equilibrium occur, the densities can be obtained from particle balance equations. Depending on the species type and the nature of the plasma, the diffusive species flux densities can be obtained from Fick’s law (1855) [32, 33], the Stefan–Maxwell equations (1867) [34] or from the drift–diffusion equation due to Townsend (1899) [35, p 193, 36].

- For stronger deviations from equilibrium, the velocity distributions of the electrons—and possibly other species—are no longer Maxwellian. Instead it is obtained from a kinetic equation, usually the *Boltzmann equation* (1872) [37].

Such equations are also needed to *derive* the expressions for the transport fluxes and the associated transport coefficients, also when the plasma is Maxwellian.

In this exposition we have deliberately been very precise in citing the original works in the respective fields. This demonstrates the fact that the theoretical basis on which plasma science could build was laid in the 19th century. Obviously, the later developments would not have been possible without the advent of ‘modern’ physics, which teaches us, among other things, the nature of radiation, the structure of the atom and the transitions between its states. That said, it is remarkable that the concepts mentioned above are still being used—in many cases without essential alterations.

One of the major challenges in plasma physics is that it has to deal with all these (classical) concepts *simultaneously*. Moreover, the transport properties and source coefficients are much stronger functions of the primary variables and feature larger gradients than one usually encounters in computational fluid dynamics, say. Toss in the necessity to deal with the atomic and quantum physical aspects and with radiation, and the complexity becomes clear. The result is that a full treatment of a plasma on the basis of first principles is feasible neither analytically nor numerically, not even with the computing power that is available today. All present plasma models are therefore compromises, judiciously reduced problems, which

nevertheless try to capture all the physical phenomena that one wishes to understand.

With the advent of electronic computing, it became possible to integrate problems that were previously too difficult to handle by other means. To some extent the increased sophistication of numerical results obtained since then can be attributed to the fact that higher processing speed and bigger computer memory have become available over the years. This is only part of the story. It would not be fair to disregard the tremendous progress in scientific computing. Examples are the development of efficient algorithms such as Krylov subspace methods [38] for iteratively solving systems of equations, and the development of advanced discretization methods such as [39] an alternative for the traditional exponential scheme [40]. Also the importance of developments in software engineering can hardly be overestimated. Concepts such as *design patterns* [41] aid the task of designing complex software, while new high-level computer languages such as C++ [42, 43] have facilitated its implementation. It is only fair to say that the *combination* of the progress in plasma physics and scientific computing and the availability of massive storage and calculation speed has brought numerical plasma physics to where it stands today.

3. Electron transport in plasmas

It would be tempting to adopt 1928 as the year in which plasma science commenced, since in that year the term ‘plasma’ was coined by Irving Langmuir [44]. That, however, would ignore the major achievements of the first three decades of the previous century. These were enabled by the better understanding of the structure of matter, which was boosted by the discovery of the electron by Thompson in 1879.

Early work on the distribution function of electrons was carried out in relation to studies of the motion of electrons in gases in the presence of electrostatic fields. Starting from the year 1899, Townsend [35, 45–47] carried out experiments and, using Maxwell’s diffusion theory [34], he derived equations that describe the variation of the partial pressure of the plasma species in the direction of the electric field. Essentially, these results [35, p 193] express the drift–diffusion equation, which is still omnipresent in gas discharge modelling. At the same time, Pidduck [48] studied the ‘abnormally high’ electron temperature in gases, see also his later historical account of that early work [49].

Some years later, but apparently unaware (see [50]) of Townsend’s work, Hertz [51] and Bartels [52] developed equivalent theoretical models, that do not depend on Maxwell’s work. Using Hertz’ theory, Druyvesteyn [53–56] derived an analytical result for the electron velocity distribution function for the case that the momentum increase due to the electric field is balanced by the reduction due to elastic collisions with a frozen background gas. In the same years, Davydov [57] demonstrated the correction that is needed if the background gas has a non-zero temperature, and Davydov [57] and Druyvesteyn [58, 59] derived expressions for the distribution function in the presence of inelastic processes, using idealized expressions for the excitation and ionization

functions. Another noteworthy paper from this period is due to Smit [60], perhaps the first researcher who applies numerical techniques to solve Boltzmann’s equation. In this paper he also takes the effects of electron–electron collisions into account, albeit in an approximate fashion.

Druyvesteyn’s 1936 paper marks an important paradigm shift: he does not merely calculate the distribution functions, but also obtains such coefficients as Townsend’s ionization coefficient α and compares the results ([60], figure 3) with Townsend’s experiments. Today, the calculation of source and transport coefficients is one of the primary applications of Boltzmann equation solvers.

3.1. Lorentz’ two-term expansion

The calculations of Townsend, Bailey, Hertz, Bartels, Davydov and Druyvesteyn were all based on a similar approach. Expressions for the electronic transport fluxes were combined with integral expressions for the energy loss in elastic collisions, and energy gain in an external field. They idealized the system, for example by assuming constant (energy-independent) values for the cross sections. Then, *exact analytical solutions* were obtained from such *approximate models*.

In the mid 1930s, an alternative approximate calculation procedure was introduced. Rather than obtaining exact solutions for idealized systems, people started to work on *approximate solutions for less-idealized problems*. The solution strategy that dominated the years to come was based on an expansion in *spherical harmonics* Y_{lms} or, under more restrictive conditions, *Legendre polynomials*. The latter has been introduced by Morse *et al* in 1935 [61]. Using the spherical harmonics method, the distribution function f is expanded as

$$f(\vec{x}, \vec{v}) = \sum_{l,m,s} f_{lms}(\vec{x}, v) Y_{lms}(\theta, \phi) = f_0(\vec{x}, v) + \frac{\vec{v}}{v} \cdot \vec{f}_1(\vec{x}, v) + \dots, \quad (1)$$

where $\vec{x}$ and $\vec{v}$ denote the position and velocity coordinates and θ and ϕ are the polar and azimuthal angle of a spherical coordinate system.

If the only anisotropy in the velocity is in the v_x direction, the first-order expansion contains two terms,

$$f(\vec{x}, \vec{v}) = f_0(\vec{x}, v) + \frac{v_x}{v} f_1(\vec{x}, v). \quad (2)$$

The latter result is mathematically equivalent to the perturbation method that was used by Lorentz in his treatise on electronic transport properties in metals [62]. While some authors seem to suggest that Lorentz introduced the more general spherical harmonics expansion³, careful inspection of Lorentz’ original texts ([62] and his textbook *The Theory of Electrons* [64, pp 267–274]) provides no evidence to substantiate that claim⁴.

³ See for example Morse, [62] above equation (4) or [63, p 2].

⁴ Interestingly enough, the latter textbook is most often cited in this context, while in that work the perturbation method is presented merely as a ‘note’. Citing [62] seems more appropriate here.

The 1935 paper by Morse *et al* has a remarkably modern look-and-feel. Both the presentation of the subject matter, as well as the notation are similar to what one typically finds in modern literature about the spherical harmonics expansion method. Morse's paper is somewhat restricted in the sense that it deals exclusively with stationary distributions in constant external electric fields. Furthermore, it assumes that electrons collide elastically with a frozen background gas. It contains a derivation of the Davydov/Druyvesteyn distribution, but no references are made to the works of Davydov [57] or Druyvesteyn [58, 59].

If we recall that also the development of the electron transport theory in gases was a largely duplicated effort, the conclusion seems justified that until the mid 1930s, the gas discharge researchers in the UK and the USA, on the one hand, and those in Russia and the European mainland, on the other, had limited access to each others' works.

In 1937, Allis and Allen [65] took the two-term expansion method one step further. While the Morse paper was concerned with the electron velocity distribution as such, this time also the electron diffusion coefficient and mobility were considered in terms of the expanded distribution function. In addition, the effect of a transversal magnetic field was considered in order to allow a comparison with the experiments of Townsend and Tizard [47]. In a second paper [66] Allen also calculates the electron 'temperature' and mobility in the noble gases Ar, He and Ne in Townsend-discharges, comparing the results with the experiments by Ramsauer and Kollath [67] and Townsend and Bailey [68–70]. Tonks and Allis [71] and Tonks [72] refined the two-term approach in the presence of crossed magnetic fields further in two subsequent papers.

The next major contribution was made by Holstein [73] in 1946. In the cited paper he considers the equation for the distribution function in high-frequency (HF) fields. After a general exposition of the influence of time-dependent fields on the distribution function, he provides analytical solutions based on the two-term expansion, considers the influence of inelastic collisions for energy-independent cross sections in detail and discusses the relation between the HF and dc results. A few years later, Barbiere uses Holstein's technique for calculating the distribution function, drift velocity and temperature of electrons in helium argon discharges [74].

An interesting account of these first decades of gas discharge research is given by Loeb in his Kelvin Lecture on April 24, 1947, see [75]. In this paper (page 230) he discusses the development of Boltzmann equation solvers and expresses his belief that the 'laborious and intricate (numerical-JvD) method of calculation initiated by Smit', [75] will be the only path forward. Since at present the Boltzmann equation is usually tackled by numerical computing, we can say that Loeb was right, although the advent of electronic computing has made the task less laborious than it used to be.

Since the subject of the theory and evaluation of the Boltzmann equation is highly theoretical in nature, it can easily be considered to be an 'academic' task. At this point it is therefore interesting to observe that the development of the solution strategies that we have discussed so far has been largely influenced by industrial research:

Hertz, Druyvesteyn and Penning were at Philips Research Laboratories in Eindhoven, Tonks and Langmuir at General Electric in Schenectady, NY, while Holstein and Barbiere worked at Westinghouse Research laboratories in Pittsburgh. This strong involvement of industry in plasma science has remained until today.

The state of the art of the transport theory of electrons and ions in electromagnetic field in the mid 50s is covered well by Allis contribution to the *Handbuch der Physik* [76], which is still regularly referred to by authors in the field.

3.2. Obtaining transport coefficients from experiments

Until the early 1960s, in most papers the Boltzmann equation was solved to obtain values that could be compared directly with experiments, such as the drift velocity or Townsend's α coefficient. The cross sections that are needed in those studies were either assumed or taken from earlier experiments. This situation changed with the major contributions of Frost and Phelps *et al*, who introduced a technique in which cross sections are *determined* using the Boltzmann calculations. In an iterative process, they adjusted the intermediate cross section values such that the converged result matches the experimentally obtained measurable macroscopic parameters as a function of the electron energy. This way they obtained electronic cross sections for momentum transfer and rotational excitation in hydrogen and nitrogen [77] and the momentum transfer cross section in the noble gases He, Ar, Kr and Xe [78]. In addition, inelastic cross sections were obtained in hydrogen and deuterium [79]. At the end of the 1960s, important contributions were made by Parker and Lowke [80] and Skullerud [81]. Using theoretical and numerical methods, they explained the anisotropic nature of electron diffusion in the presence of electric fields, showing that the longitudinal diffusion coefficient is considerably smaller than the transversal diffusion coefficient, consistent with experimental results obtained a few years earlier. In their 1974 textbook 'The Diffusion and Drift of Electrons in Gases', Huxley and Crompton provide a comprehensive discussion of the state of the art in the field at that time. In addition, it contains a highly interesting historical account of the early work on early experiments on electron transport.

In the 1970s and 1980s, Wilhelm and Winkler published an impressive series of papers about the theory, implementation of the two-term approximations and its application in self-consistent plasma simulations. They studied the influence of inelastic processes on the calculation of transport coefficients [82, 83], thereby providing a detailed analysis of the collision integrals [84, 85], calculated the electron energy distribution function (EEDF) and transport coefficients in helium–mercury discharges [86] and investigated the relaxation of the EEDF in response to stepwise changes in the electric fields [87–91] and the transient behaviour of the EEDF in HF discharges [92, 93]. In a series of two papers they provided a—for that time—highly sophisticated self-consistent model for fluorescent lamps [94, 95], which includes a calculation of the (non-Maxwellian) EEDF.

3.3. Limitations of the two-term expansion: higher-order approximations

While the limitations of the two-term approximation have been clear from the start, higher-order approximations were not developed until the early 1970s. A three-term approximation was developed by Francey and Stuart [96], who established criteria for the validity of the two-term approximation and investigated the influence of higher-order terms on the calculation of transport coefficients. Similar work was done by Ferrari [97, 98], who also discusses the calculation of the collision operator in the second-order approximation in more detail. The insufficient accuracy of the two-term approximation was noted by other authors as well, like Makabe *et al* who compared theoretical results with experiments in argon [99] and xenon [100]. Also Makabe later provided a three-term implementation [101].

In 1980, Braglia [63] discusses the limited validity of the two-term approximation, but expresses his opinion that the extensions to higher orders result in equations that are too complex to be of practical use. He also claims that for ‘a number of well-known reasons, Monte Carlo (MC) methods will never become a convenient tool for analysis and systematic calculations’. Braglia states that an integral formulation of the Boltzmann’s problem may be promising. Such formulation, he argues, was developed already in the early days of discharge physics, but seems to have been forgotten ever since.

In spite of Braglia’s skepticism, an extension of the expansion technique to arbitrarily high orders was presented one year later by Pitchford, O’Neil and Rumble [102]. It contains a thorough analysis of the order that is needed to obtain expressions for the transport coefficients that are sufficiently accurate for the purpose of reconstructing cross sections from swarm data. For nitrogen discharges they showed that four to eight terms need to be taken into account. This application to nitrogen plasmas is elaborated on in a subsequent paper [103], which also contains a quantitative analysis of the errors made when the two-term expansion is used instead. In this work, a steady state is assumed and no ionization processes have been taken into account, limiting the approach to moderate reduced field values. Both papers provide important information about the numerical procedure, such as the energy mesh and boundary conditions⁵, which commonly get less than appropriate attention in papers about this subject.

In 1989, Winkler *et al* [104] presented results of a multi-term expansion that handles non-conservative electron collisions, such as attachment and ionization as well. In this work, three key assumptions are made: the plasma is uniform, scattering is isotropic and the electrons equally share the energy available after an ionization event has occurred. Later, Loffhagen presented a non-stationary treatment of the Boltzmann equation: [105] discusses the two-term expansion, discussing the numerical procedure and discretization issues with a level of detail which is, unfortunately, not matched by many other papers about the subject. In [106] a generalization to multi-term expansions is discussed.

The higher-order techniques that we have mentioned so far were mostly rather straightforward extensions of the two-term approximation that was initiated by Lorentz and, in the context of gas discharge physics, Morse, Allen and Allis. A more fundamentally different approach was developed by Lin *et al* [107], based on earlier work on ion transport by Kihara [108] and Viehland and Mason [109, 110]. In this paper, the coefficients f_{lms} that occur in the spherical harmonics expansion (1) are further expanded in terms of Sonine polynomials ([107], equation (14)). Although this further complicates the mathematics, the resulting expansion coefficients have a clear physical meaning, representing such concepts as the drift and transversal velocities ([107], table 1). In the mathematical procedure, Boltzmann’s equation is rewritten as an (infinite) set of coupled *moment equations*, the solutions of a truncated system can be used to obtain the electron transport coefficients. This method avoids the calculation of the EEDF itself. The procedure is discussed and extended to include non-conserving reactions in a number of later papers, see, for example, [111, 112].

3.4. Lessons learned, practical considerations

Since the work of Lin, Robson, Ness and White is probably the most rigorous and complete treatment of electron transport that we have at present, one could expect that Boltzmann equation solvers that are based on this approach are the *de facto* standard nowadays. That turns out not to be the case. Many—if not most—authors use variations of the traditional two-term expansion method in their swarm and discharge models. In a 2003 paper White *et al* [113] make some critical comments on that practice. We summarize these as follows:

- **Mathematics:** commonly, authors claim that the two-term approach does a good job, without defining quantitatively what ‘good’ means. The accuracy of the two-term truncation should be tested: that can be done by comparing the results with higher-order techniques or by calculating an upper bound on the error by other means.
- **Physics:** typically, assumptions are made in the physical model of the problem at hand. As an example, White *et al* mention the choice for the way the non-conserving processes are dealt with ([113], section 3.2.1). Mathematical results can never be more accurate than the physical model for which they are obtained.
- **Context:** even if we have an objective measure of the accuracy of a physical–mathematical model, the applicability of that model depends on the context in which it is used. As an example, when Boltzmann calculations are used for the analysis of swarm experiments, a much higher accuracy is needed than is necessary for the more straightforward calculation of source and transport terms for known or assumed cross section data.

White *et al* conclude that, for these and other reasons, multi-term analysis should be the norm, rather than the exception. His major recommendation is that ‘researchers in low-temperature plasma physics spell out assumptions made in their models, and present error bars with their results.’ This suggestion is as non-controversial, as it is embarrassing for some.

⁵ See, for example, notes 19–21 in [103].

3.5. Papers related to electron transport in this cluster issue

A number of papers that are part of the present cluster deal with the Boltzmann equation and related topics. White *et al* [114] have provided an overview of recent developments in the study of the Boltzmann equation in connection with the fluid equations for the modelling of charged particle transport in plasmas.

Petrović *et al* contributed a paper [115] about the analysis of data obtained in swarm experiments. In addition, it extensively discusses how these data should be used in a fluid model. This is an important issue, since after all, transport coefficients have a meaning only in conjunction with the equations in which they occur.

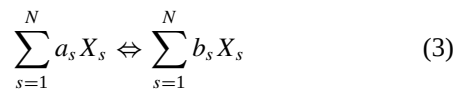
The contribution of Mason [116] deals with ‘The Status of the Database for Plasma Processing’. He discusses current efforts to establish a novel repository for input data that are relevant for the plasma (modelling) community. Since we have seen that Boltzmann-based methods are a key ingredient in the post-processing of experimental data, it is appropriate to announce that paper in the present context.

The electronic Boltzmann equation plays a central role in two other papers in this cluster. Both papers employ mathematical techniques to discuss particular problems in radio-frequency discharges. In the contribution of Turner [117], the concept of ‘collisionless heating’ is reviewed, while the paper of Brinkmann [118] discusses analytical methods for accurately describing the plasma sheath in such discharges for a wide parameter range.

4. Plasmas in or near LTE

4.1. The plasma composition

Under conditions of LTE, the composition of a gas mixture is governed by the local pressure and temperature. The *mass action law* establishes expressions for the ratios between the densities of the particles. For a mixture with N species X_s , a chemical reaction can be written as



and the mass action law can be stated as

$$\prod_{s=1}^N n_s^{(a_s - b_s)} = f(T), \quad (4)$$

where $c_s \equiv a_s - b_s$. The function $f(T)$ depends on the nature of the reaction and can be expressed in terms of the species’ *partition functions* [119, p 203]. For excitation, ionization and association reactions, the equation reduces to the well-known Boltzmann, Saha and Guldberg–Waage relations, respectively.

For a system of N_s species, composed of N_α different elements, the mass action law establishes $N_s - N_\alpha$ independent equations. In order to solve the system it must be augmented by α equations that prescribe the numbers of atoms of each

element per unit of volume. These *elemental densities* of elements α are given by

$$n_\alpha = \sum_s R_{s\alpha} n_s, \quad (5)$$

where the sum goes over all species, and $R_{s\alpha}$ is the number of atoms of type α in the species s . Depending on the system, equivalent constraints can be provided, for example by specifying a set of partial pressures in the system. If, as in a plasma, charged species are present, an additional constraint is imposed by the condition of (quasi-)neutrality, $\sum_s n_s q_s = 0$.

Nowadays, the mass action laws (including the precise form of the reaction-specific function f_s) are usually obtained as a result of a variational problem, by minimizing the Gibbs energy of the mixture. Indeed, it is also possible to obtain the species composition directly from a variational formulation of the problem. However, such a procedure was deemed impractical in the pre-computer era, and the relation between the mass action law and Gibbs’ energy was considered to be interesting mostly from a theoretical point of view until the 1960s of the previous century. A detailed discussion can be found in Shapiro and Shapley [120]. The cited work contains a rigorous mathematical discussion of the properties of the mass action laws, their equivalence to the variational formulation, as well as proofs of existence and uniqueness of solutions.

Obviously, this discussion of LTE phenomena is not restricted to the plasma state: it applies equally well to other multi-component equilibrium gas mixtures. However, in plasmas some particular additional difficulties are encountered. Griem’s *Principles of Plasma Spectroscopy* [119, chapter 7] provides an excellent, self-contained discussion of the thermodynamic equilibrium relations for plasmas. Another great discussion of the application of the mass action laws in high-enthalpy mixtures can be found the textbook of Zel’dovich and Raizer [121, p 176]. The latter work also mentions a 1938 publication of a proof of existence of a unique solution ([6] in [121], p 191), apparently not known to Shapiro and Shapley.

Firstly, in a plasma at not too low temperatures, excited states need to be taken into account in the calculation of the partition function. In principle, the number of excited states is infinite, and although the population decreases for higher quantum numbers, it can easily be shown that the growth of the statistical weight makes the internal partition sums diverge [122]. That analysis, however, is based on an isolated atom. In atoms in a plasma, the number of bound states is finite, limited by the Fermi and Griem criteria. The first criterion excludes states with electrons at radii bigger than the (mean) inter-atomic distance ($1/\sqrt[3]{n}$), the second takes into account the ionization energy lowering due to the influence of the plasma potential [123, p 131]. For an alternative formulation, see [124, p 139], more details can be found in [119, section 7.4]. As a consequence, the partition functions are no longer functions of the temperature alone: additional dependences on the pressure and the densities of charged species are induced.

In addition, deviations from the ideal gas may need to be taken into account. The *Debye–Hückel* theory predicts a pressure reduction, which is given by

$\delta p_{\text{dh}} = kT/24\pi\lambda_{\text{d}}^3$ [123, equations (14) and (15)], where λ_{d} is the Debye length. This may alternatively be written as $\delta p_{\text{dh}} = n_e kT/18\Lambda \approx \alpha p/18\Lambda$, where α is the ionization degree and Λ is the *plasma parameter*, the number of electrons in the Debye sphere. We see that the relative pressure correction depends on the ionization degree and plasma parameter as $\delta p_{\text{dh}}/p \approx \alpha/18\Lambda$. We see that the pressure correction is usually small, especially for *ideal plasmas* for which $\Lambda \gg 1$.

Recently, Collona *et al* have developed a novel way of calculating the equilibrium composition of plasmas in complex gas mixtures, based on earlier work by Villard (see [125, 126] and references therein). In these iterative algorithms, a *distance* is associated with each mass action law, which is a measure for the deviation from equilibrium, given the intermediate composition result. Essentially, the composition is updated to continuously make the largest distance zero, hereby modifying other distances. A convergence criterion can be based on the smallest distance in the set. Collona's improved algorithm reportedly has extremely good convergence properties, and does not require human intervention to circumvent convergence problems, unlike Villard's earlier version.

4.2. Transport coefficients

Now that we have discussed some issues involved in the calculation of the composition of equilibrium plasmas, let us turn our attention to the calculation of the *transport coefficients*. We are obviously not able to cover that subject in any depth, but will illustrate the key issues by looking at a particular example, the calculation of the coefficient of thermal conductivity.

In a flowing gas mixture, heat can be transported by means of *conduction* and *material transport*. Using Fourier's heat conduction law, we find that the heat flux density $\vec{q}$ can be expressed as

$$\vec{q} = -\lambda_{\text{f}} \vec{\nabla} T + \sum_{i=1}^{\mu} h_i \rho \vec{V}_i, \quad (6)$$

where T is the temperature and λ_{f} is the *frozen* heat conductivity, to be used in the absence of chemical reactions. The rigorous calculation of the frozen transport coefficients is a formidable task, which is usually carried out using the formalism of Chapman–Enskog [127, chapter 8]. We refer to Devoto [128] for an early application of that technique to partially ionized gases.

Let us now consider the contribution to the conductivity due to material transport. In the last term of the previous equation, the sum runs over all species and h_i and $\vec{V}_i$ are the specific enthalpies and average velocities of species i , respectively. If we decompose the latter in *barycentric* and *diffusive* parts $\vec{V}$ and $\vec{W}_i$ and re-order terms, we get

$$\vec{q} = h\rho \vec{V} - \lambda_{\text{f}} \vec{\nabla} T + \sum_{i=1}^{\mu} h_i \rho \vec{W}_i. \quad (7)$$

The first term represents the *convective heat flux*, where $h = \sum h_i$ is the specific enthalpy of the mixture; the last expresses

the heat flux associated with the diffusive motion of species. Under LTE conditions, Butler and Brokaw [129] arrived at the result

$$\vec{q} = h\rho \vec{V} - (\lambda_{\text{f}} + \lambda_{\text{r}}) \vec{\nabla} T. \quad (8)$$

The expression for the *reactive* thermal conductivity λ_{r} turns out to depend on the temperature, the species' enthalpies and molar fractions and the coefficients of binary diffusion of all species pairs.

Later works, most notably those of Murphy *et al* [130], Rat *et al* [131], Magin and Degrez [132] and Rini *et al* [133], further developed the transport theory in (near-) LTE plasmas, in particular by considering the diffusive fluxes and associated transport coefficients with a higher level of sophistication. Calculations of transport coefficients in LTE plasmas have been carried out by numerous other authors. Among the most thorough studies in the last years are those performed in Bari by Capitelli *et al* [134, 135] and Bruno *et al* [136–138]. In these papers it has been shown that the influence of the excited states, with their 'abnormally high' diffusion cross sections is significant in hydrogen plasma. Collona's composition calculation algorithm (see above) has recently been used by D'Angola *et al* to evaluate the thermodynamic and transport properties of air under equilibrium conditions for a wide parameter range [123]. We also mention a similar recent study by Sourd *et al* [139] for a nitrogen plasma.

4.3. Elemental diffusion

In many plasma sources, it is reasonable to assume that the pressure fractions of the various elements are uniform throughout the plasma. In such cases the pressure fractions usually follow from the specified composition at the inlet in an open geometry. For closed vessels, such as those of plasma lamps, the fractions may result from the atom contents or, in the case of saturated species, be deduced from the temperature of the coldest spot on the vessel wall.

There are also situations where the composition can be obtained from the mass action laws, while the elemental densities (equation 5) are *not* uniform. In this case we speak about *segregated (LTE) mixtures*. As an example, such plasma is formed in metal–halide lamps. In such systems the interplay between convection and diffusion results in *radial* and *axial* segregation [11, 140–143]. In this case, the elemental pressures can be determined from a set of *elemental transport equations* [11, 144], one for each element present in the plasma.

We will briefly sketch the procedure here. Let us consider a stationary plasma, consisting of species s . The relation between each species' flux density $\vec{\Gamma}_s$ and its net rate of production S_s is governed by the usual balance equation; in flux-divergent form it reads

$$\vec{\nabla} \cdot \vec{\Gamma}_s = S_s, \quad (9)$$

where $\vec{\Gamma}_s$ and S_s are the particle flux density and volumetric production rate of species s , respectively. As before we write the number of atoms of element α in the species s as $R_{s\alpha}$, then the *elemental* flux density is given by

$$\vec{\Gamma}_{\alpha} = \sum_s R_{s\alpha} \vec{\Gamma}_s. \quad (10)$$

Since—unlike species—atoms are conserved in reactions, we arrive at the elemental continuity equation,

$$\vec{\nabla} \cdot \vec{\Gamma}_\alpha = 0. \quad (11)$$

The species fluxes are governed by convection–diffusion equations [11]. Substitution of those expressions in equations (10) and (11) yields the convection–diffusion equation for all elements; these can be solved to obtain the elemental pressures. Together with the local temperature, these can be used with the mass action laws to obtain the species' composition.

The details of this algorithm, including the expressions for the effective convection pseudo-velocity and diffusion coefficient, are discussed in recent papers by Hartgers *et al* [144] and Beks *et al* [141]. This procedure has been implemented in the Plasimo framework as part of an effort to model axial segregation in metal–halide lamps [142, 143].

4.4. Papers about (near-) LTE modelling in this cluster issue

We welcome three contributions about the modelling and numerical simulation of (near-) LTE plasma in this cluster. Recent developments in the modelling of transport in (near-) equilibrium flow with high Mach-numbers are reported in the contribution of Degrez *et al* [145]. In the contribution of Capitelli *et al* [146], the transport coefficients in thermal plasmas are considered, special attention is paid to the influence of the presence of electronically excited states. The modelling of thermal plasmas—and in particular the diffusive fluxes—in a thermal plasma suitable for arc welding is described in the contribution of Murphy *et al* [147].

5. Particle and hybrid modelling techniques

5.1. Limitations of continuum models

The Boltzmann-based methods (section 3) and the LTE methods (section 4) yield results which are continuous with respect to the space coordinates. Those equations describe the averaged statistical properties of the plasma particles—such as the species densities, velocities and mean energies (temperatures)—rather than their individual properties. These methods are therefore referred to as *continuum* or *fluid* descriptions. Both terms can be used interchangeably, although the latter has the disadvantage that the word fluid can be erroneously understood to have its more restricted meaning of liquid⁶.

Such continuum description breaks down for highly diluted plasmas, or for species with extremely low abundances. It is also impractical or inaccurate in situations where the deviations from equilibrium are strong, for example due to the presence of large gradients or strong transient effects. Such systems are therefore better dealt with using *particle methods*, such as the MC techniques. One must be careful with the alternative term *kinetic methods* because it seems to have various connotations: a quick survey on the Internet revealed

that it is sometimes used as a synonym for particle methods, although the majority viewpoint seems to include statistical (Boltzmann-based) methods as well in the definition.

5.2. Particle modelling

In the MC method, a collection of particles that represent the species population are followed in space and time. The stochastic nature of elastic collisions and inelastic collisions in which particles are produced or destroyed is emulated by letting such processes happen at random times, but with an average frequency which matches the real frequency. The method relies on relatively elementary concepts, such as Newton's laws for particle motion and a specification of the chance of interactions to happen.

In order to ensure that a good coverage of the coordinates in configuration and velocity space is obtained, it is required that a sufficiently high number of particles is tracked. At present, numbers of up to hundreds of million are not unusual. It is obvious that a table of the positions and velocities of that many particles is not fit for human consumption. Indeed, a MC run is usually followed by a post-processing stage in which averaged properties are reconstructed from the raw population data, usually as a function of position.

The MC method raises an interesting issue about the role of modelling in the quest for better understanding plasma physics. On the one hand, of the techniques we have discussed so far, the MC technique is closest to reality, since it mimics nature more directly than continuum methods do. On the other hand, this does not imply that MC provides most *insight* into how a system works. Indeed, insight⁷ into the predominant characteristics of a system is not obtained by the construction of an exact copy, but rather by the realization of an appropriate abstraction of that system. In view of these observations, it may be more appropriate to think of MC methods as *simulation techniques*, rather than *models*.

5.3. Molecular dynamics

In recent years, *molecular dynamics* studies have provided a new means for studying particle behaviour on the molecular scale. The method is capable of handling matter in the condensed state, which makes it the method of choice for studying plasma–wall interactions. Obvious applications are concerned with film growth, etching and plasma sputtering.

Although the method celebrated its 50th birthday last August [148], it has been widely adopted only recently. This can be attributed to the fact that the method is computationally highly demanding, so commodity computer hardware used to be insufficiently powerful for this type of simulation until a few years ago. We refer to [149], part of this cluster issue, for an extensive discussion of this exciting, powerful technique.

5.4. Hybrid modelling

A MC model results in the positions and velocities of a representative set of particles, from which the usual quantities

⁶ This problem is not restricted to the English language. For example, in Dutch, the word *vloeistofmodel* (liquid model) is commonly used where the more general concept of a *continuümmodel* or *fluidummodel* is meant.

⁷ Merriam–Webster: *Insight* (2): the act or result of apprehending the inner nature of things or of seeing intuitively.

of interest, such as densities, velocities and mean energies can be obtained directly. Alternatively, the MC technique can be used to obtain the transport and source coefficients of one or more of the plasma species. These are then used in a continuum model, which calculates the primary plasma variables. In the same way, a Boltzmann solver can be used to handle the non-equilibrium features of the plasma under study and combined with a fluid model. Such approaches, where various methods are combined to simulate the plasma as a whole are known as *hybrid modelling techniques*.

Early hybrid simulations used multiple models for the electron gas. These modelled the low-energetic electrons with a fluid approach, while the highly energetic electrons, capable of exciting and ionizing heavy particles were simulated with the MC technique. The first such study was carried out by Belenguer and Boeuf in the early 1990s [150]. This particular combination of models is referred to as *beam-bulk* modelling in the later literature [151], although this term seems to have been mostly abandoned more recently.

Other hybrid models combine the MC technique with the particle-in-cell (PIC) method. This particular type of calculation is thoroughly explained in the textbook of Birdsall *et al* [152]. Essentially, in this method the individual particles are tracked on a continuous mesh, while their population-averaged local properties and the electromagnetic field are represented on a fixed discrete mesh [153–155]. A noteworthy descendant of this method is the *accelerated* PIC scheme due to Schweigert *et al* [156, 157]. This method speeds up the determination of the periodic steady state in RF plasma reactors [158] by acting appropriately on the knowledge that the electronic and ionic transport differ by orders of magnitude. Other important efforts to mitigate the computational costs of the method were undertaken by Kawamura *et al*, see [159].

Hybridization can also be achieved by treating different parts of the plasma *geometry* in different ways. Such *spatial hybridization method* has recently been developed by Li *et al* [160] for the study of the propagation of streamer discharges. This work combines the MC technique for the non-thermal electrons in the streamer head with a fluid model for the species motion in its tail. The latter sub-model uses the adaptive grid refinement technique that had been developed a few years earlier by Montijn *et al* [161]. Li's paper (and the references therein) provides a detailed discussion of the practical issues involved in connecting the kinetic region in the streamer head with the fluid region in its tail and represents the state of the art in streamer modelling.

5.5. Papers about kinetic and hybrid modelling in this cluster issue

In this special issue, we welcome various contributions that cover recent theoretical developments and their practical applications in the field of kinetic and hybrid modelling.

A topical review by Graves and Brault [149] deals with the subject of molecular dynamics. After discussing various models for the inter-molecular potentials, it demonstrates the application of the technique to plasma–surface interaction studies. The focus is on the simulation of the growth of thin silicon and carbon films.

A general introduction to hybrid modelling has been provided by Kushner. His paper [12] discusses how the task of self-consistently modelling low-temperature plasmas can be decomposed into a set of sub-tasks, and introduces the various algorithms for carrying out these tasks that have been implemented in the HPEM, which has been developed by the author and his co-workers. The paper demonstrates the application of HPEM to low-pressure inductively coupled reactor plasmas.

In the same spirit, the contribution by Van Dijk *et al* [14] describes the various options for simulating plasmas that are available on the Plasimo modelling platform. The paper describes the capabilities and limitations of the individual modules, and discusses how these have been deployed in previous simulation studies.

A contribution by Guo *et al* [162] discusses the application of the MC technique for the time-resolved simulation of profile etching. The three-dimensional nature of the model allows the simulation of such features as surface roughening, that could not be understood from a two-dimensional model.

Goedheer *et al* [163] discuss the modelling and simulation of dusty plasmas under micro-gravity conditions. It introduces the contemporary models for the charging and the dynamics of the dust grains and demonstrates results of a continuum model as well as a (hybrid) model in which a MC method for the electron gas is combined with the PIC technique for calculating the electrostatic field in the plasma.

Ferreira *et al* [164] have contributed a paper about the modelling of non-equilibrium argon–nitrogen microwave plasmas. It contains an extensive discussion of the calculation of the electromagnetic field for surface discharges with various geometries. In their hybrid models, fluid equations are solved self-consistently with the two-term approximation of the Boltzmann equation (see section 3.1).

Bogaerts *et al* [165] discuss the hybrid modelling of a magnetron sputtering device. In such a device, the outward motion of the electrons is impeded by a magnetostatic field. The presence of such a field clearly complicates the modelling and simulation of such a discharge. The paper discusses various hybrid schemes, which include fluid, MC, PIC and analytical ingredients. The pros and cons of the various approaches are clearly pointed out.

Unfer and Boeuf [166] present simulation results on the plasma dynamics in a surface dielectric barrier discharge [167, 168] which is excited by nanosecond pulses. They demonstrate that under the conditions that they have studied, the plasma gas is locally heated, which gives rise to shock waves which are microscopic in size. Here we see another example of a simulation in which various plasma-physical modelling types (for the electrodynamics, the non-thermal plasma dynamics and for the flow field and background gas temperature) are coupled. It therefore makes a great example of the application of hybrid strategies to contemporary problems in plasma technology.

Hagelaar *et al* [169] discuss a model for a microwave plasma which is sustained by electron cyclotron resonance (ECR). It evaluates the electromagnetic field in the (finite difference) time domain (FDTD). Unlike previous works in

this field, it employs a pure fluid description for all plasma species and assumes a Maxwellian distribution equation for the electrons. The paper presents a detailed discussion of the semi-implicit statement of the problem, which allows larger time steps than a fully explicit scheme would. Special attention is paid to the complications which result from the facts that the plasma is magnetized and that inertia terms in the equations for the ionic species must be taken into account.

6. Concluding remarks

In this paper we have discussed the main types of plasma modelling and simulation techniques. In section 2 we have argued that the theory that underlies modern plasma simulations has its roots mostly in the 19th century achievements of Fourier, Navier, Stokes, Maxwell and Boltzmann. Plasma physics could develop in an era in which atomic and quantum physics provided additional models for the structure of electrons, atoms, radiation and their interaction.

Since self-consistent plasma models involve essentially all of these concepts, and are therefore too complex to handle even with modern electronic computing, different branches of plasma physics came into existence. These have specialized on particular types of applications or application groups, the division happened to correspond approximately to the different degrees of departure from equilibrium as well.

Yet, in the last two decades the tide seems to be turning. Those who have traditionally focused on plasma in (local) thermal equilibrium have started to worry about equilibrium deviations, see for example [11, 145]. At the same time, researchers in the field of low-pressure (reactor) plasmas have started to consider the effects of flow and gas heating [9, 12, 166], which were previously studied mostly in the context of (near-) LTE plasma science. The generally pleasant and cooperative atmosphere in the plasma physics community as a whole has ensured that knowledge quickly diffused over the traditional borders.

It can be expected that this trend will continue. More than ever, the modelling and development of new plasmas sources will be a multi-disciplinary effort with electromagnetic, flow and multiple (traditional) plasma-physical aspects. This is in particular the case for the biomedical plasma applications that are being considered at present [5, 16–20, 170, 171], since the further development and deployment of such sources require a close coordination with biologists and medical specialists as well.

Cooperation has traditionally taken place on the level of physics modelling, less on the level of the actual implementation of these ideas in computer codes. It is interesting to see whether the latter will start happening as well. This would require that people open up the source codes of their implementations. An open source model has the obvious advantages that it reduces the duplication of effort and that one can modify the codes to his or her needs. Equally important is the more conceptual issue that it would allow the process of peer review to establish the codes' tacit assumptions, correctness and accuracy. This used to be good practice in previous times (see for example [172–180]), but is less usual

nowadays. Among the exceptions are Kushner's HPEM code [12, 13] and the Plasimo toolkit [15].

There is no single path forward. In the years to come, plasma science will continue to benefit from:

- advanced work on particular plasma aspects, such as its structure, composition and transport properties (this issue: [114–118, 145–147, 149, 162, 181, 182]);
- the development of reduced models for complete plasma systems that provide 'engineering rules', which capture the essential features and trends (see, for example, the analytical models which are part of this issue [165, 183], and text books like such as [3]);
- the refinement of self-consistent plasma simulation tools that has a level of sophistication and accuracy that makes them useful as CAD-tools in plasma engineering ([12, 14, 163–166, 169]);

We hope that this cluster issue will help the reader to get a feeling for the recent developments in these fields, and that this introduction has served its purpose of clarifying how these are related.

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